B.Tech. Degree III Semester Special Supplementary Examination June 2013

SE 304 MANUFACTURING PROCESSES

(2006 Scheme)

Time : 3 Hours Maximum Marks : 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) What is malleable cast iron? Describe how it is produced.

(b) Write short note on carbon steels.

(c) What is thermit welding? Explain.

(d) Describe the oxy-acetylene flame cutting briefly.

(e) What is a gating system? Explain.

(f) List out the merits and applications of die casting method.

(g) What are the basic press operations? Explain briefly.

(h) Describe the AJM process and list its applications.

PART B

(4 × 15 = 60)

II. (a) What is solid solution? Explain the different types of solid solution with examples. (9)

(b) What is carburising? Explain the different methods briefly. (6)

OR

III. Draw the iron-carbon equilibrium diagram, mark the different temperature, composition and phases and write down the important reactions taking place. (15)

IV. (a) What are the different flames used in gas welding? Draw each flame, mark the different phases and explain the application. (9)

(b) What is friction welding? Explain with an example. (6)

OR

V. (a) What is resistance welding? Briefly explain any three types of electric resistance welding. (9)

(b) What are the common welding defects? Explain. (6)

VI. (a) What are the standard allowances used in pattern making? Explain briefly. (9)

(b) What is die-casting? Explain the process with anyone type of hot chamber die casting machine. (6)

OR

VII. With a neat figure explain the construction and working of a cupola furnace highlighting the various zones. (15)

VIII. Classify closed die forging and explain each type briefly with sketches. (15)

OR

IX. Explain the mechanical and hydraulic types of shaper drive mechanism with neat sketches. Also list out the merits and drawbacks of hydraulic drive. (15)

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